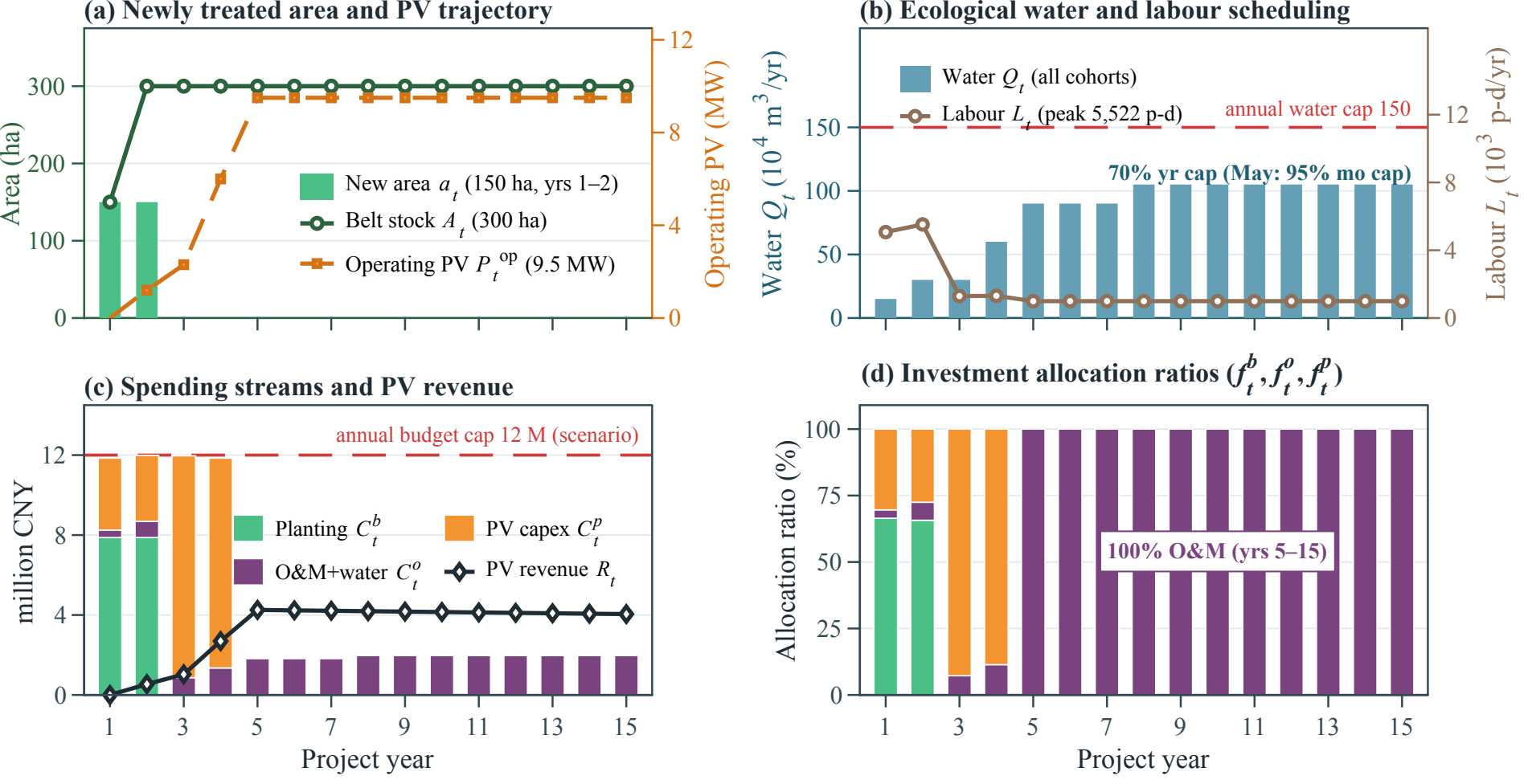

(a) Newly treated area and PV trajectory
Area (ha)
Operating PV (MW)
New area a_t (150 ha, yrs 1–2)
Belt stock A_t (300 ha)
Operating PV P_t^{op} (9.5 MW)
(b) Ecological water and labour scheduling
Water Q_t (10^4 m^3/yr)
Labour L_t (10^3 p-d/yr)
Water Q_t (all cohorts)
Labour L_t (peak 5,522 p-d)
annual water cap 150
70% yr cap (May: 95% mo cap)
(c) Spending streams and PV revenue
million CNY
annual budget cap 12 M (scenario)
Planting C_t^b
PV capex C_t^p
O&M+water C_t^o
PV revenue R_t
Project year
(d) Investment allocation ratios (f_t^b, f_t^o, f_t^p)
Allocation ratio (%)
100% O&M (yrs 5–15)
Project year